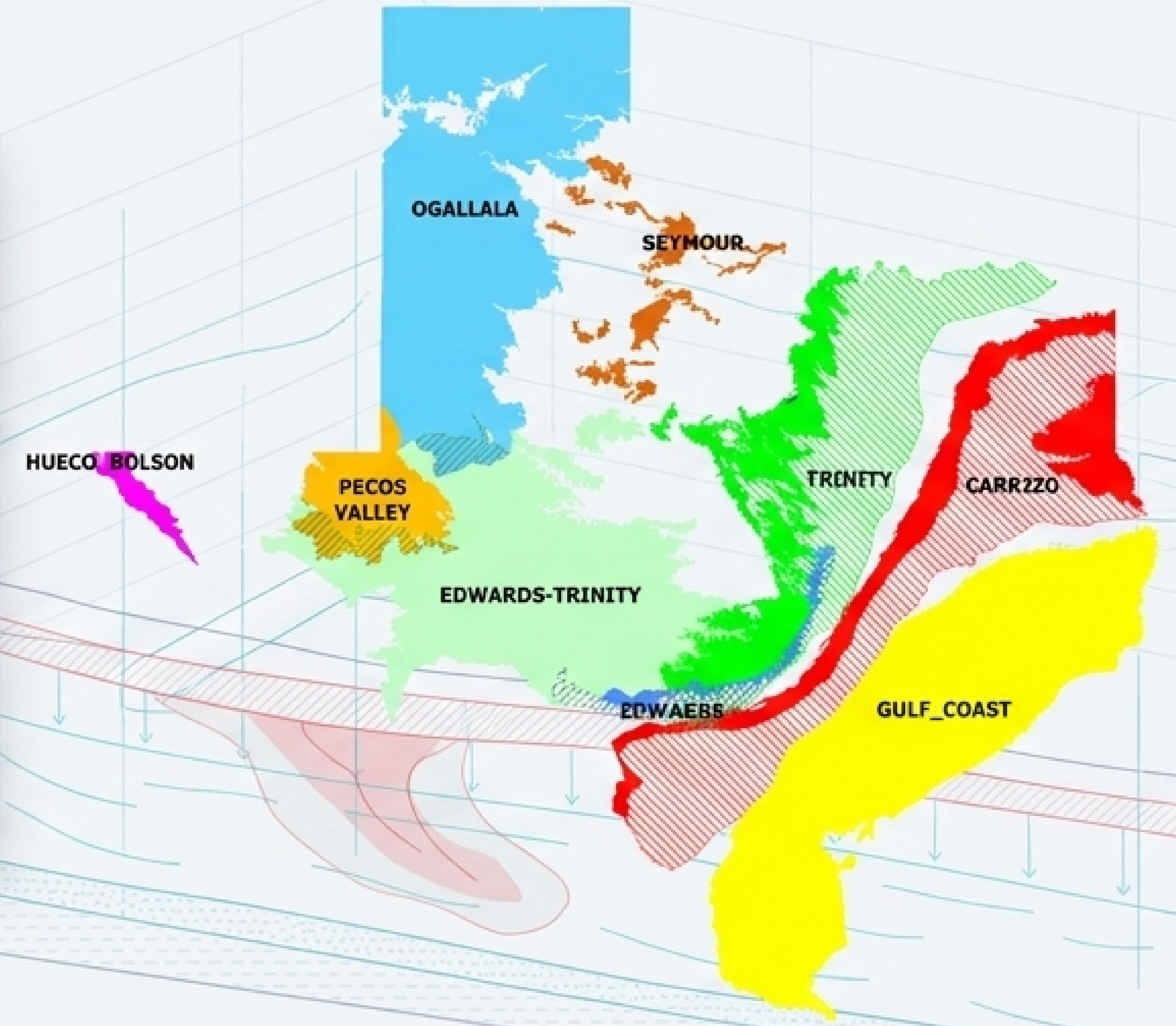
Predicting the Invisible

The Threat

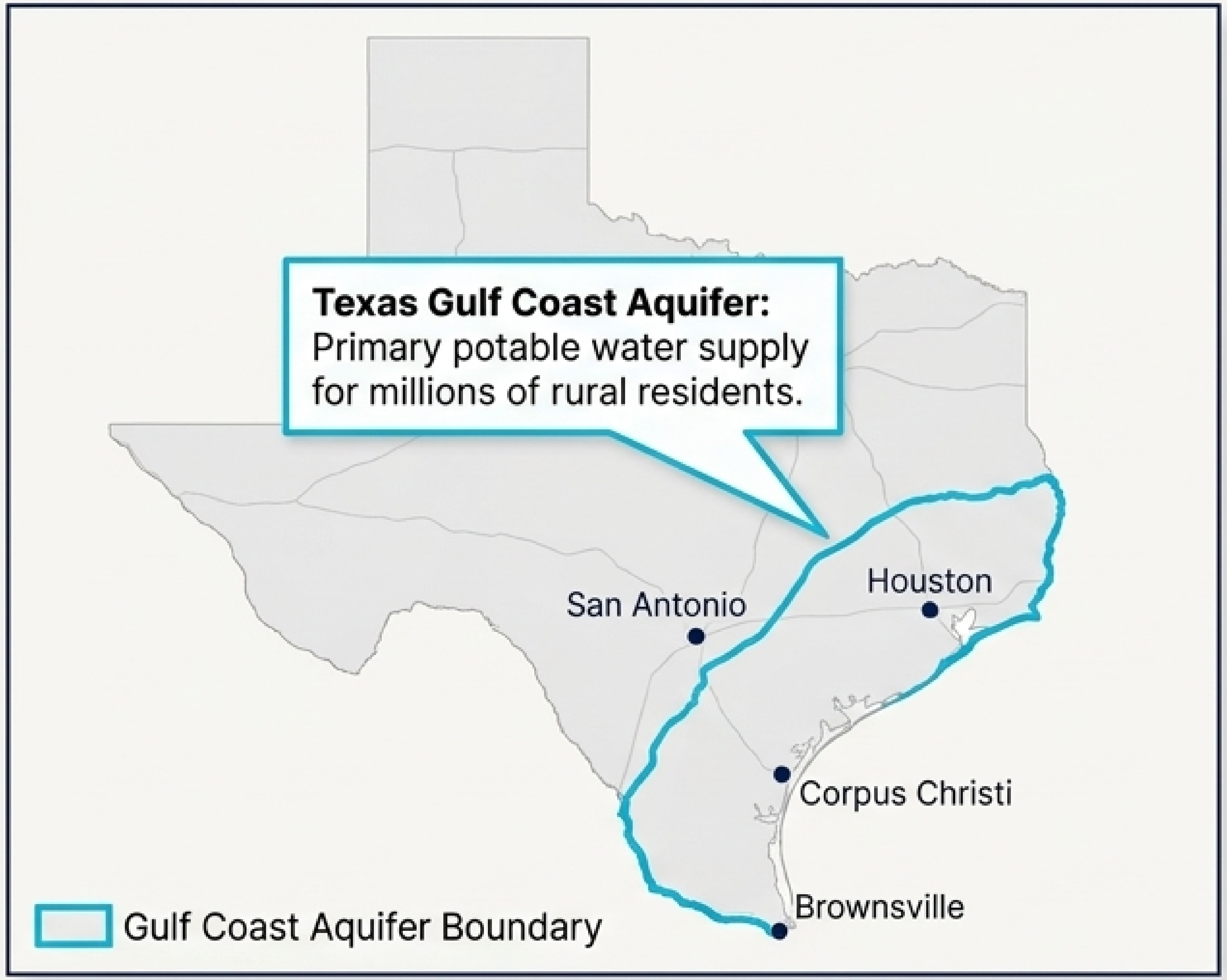
Nitrate-N: The most widely detected contaminant in global groundwater systems.

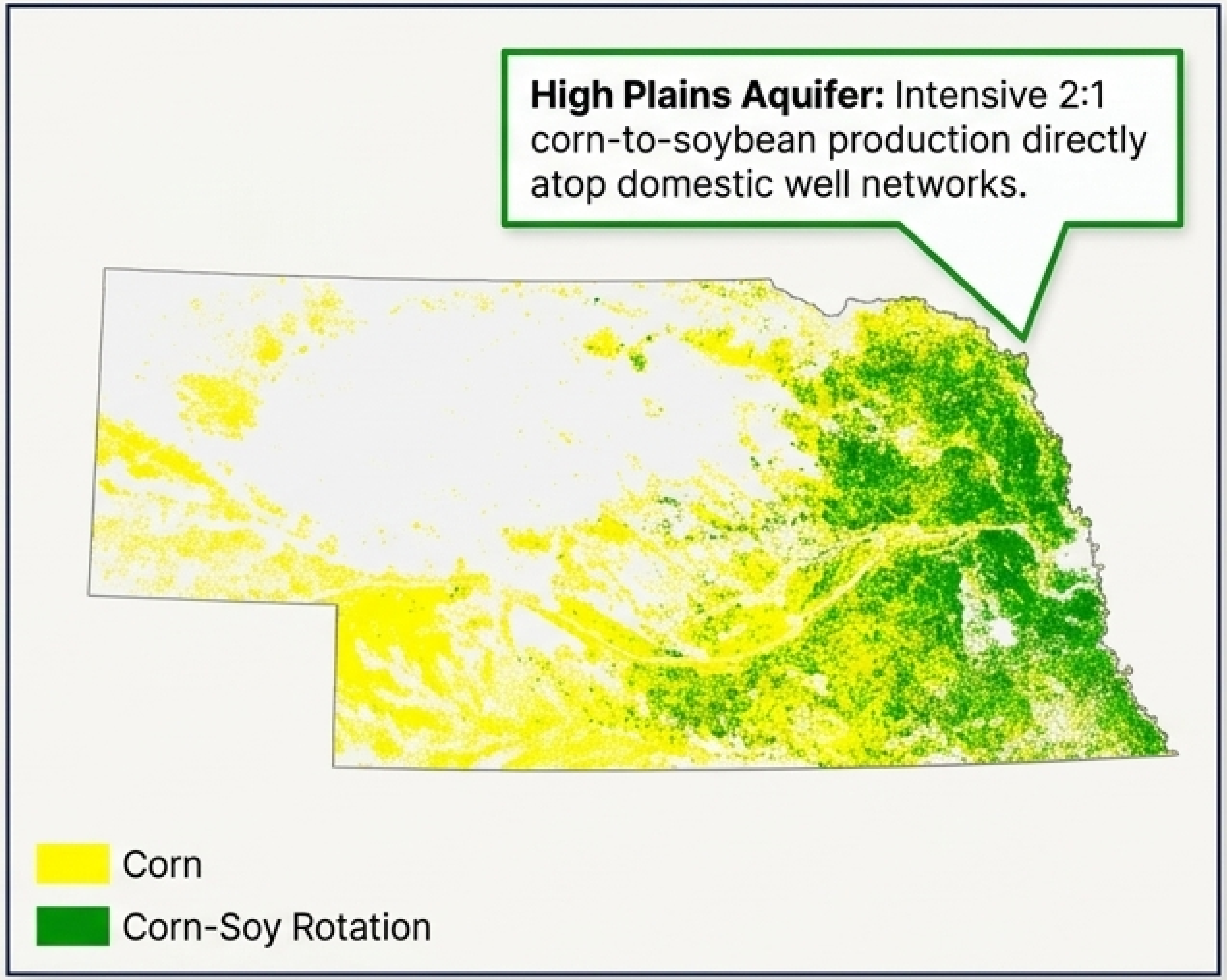


The Solution

Integrating Multiple Logistic Regression (MLR) and Life Cycle Costing (LCC) for proactive socioeconomic policy.

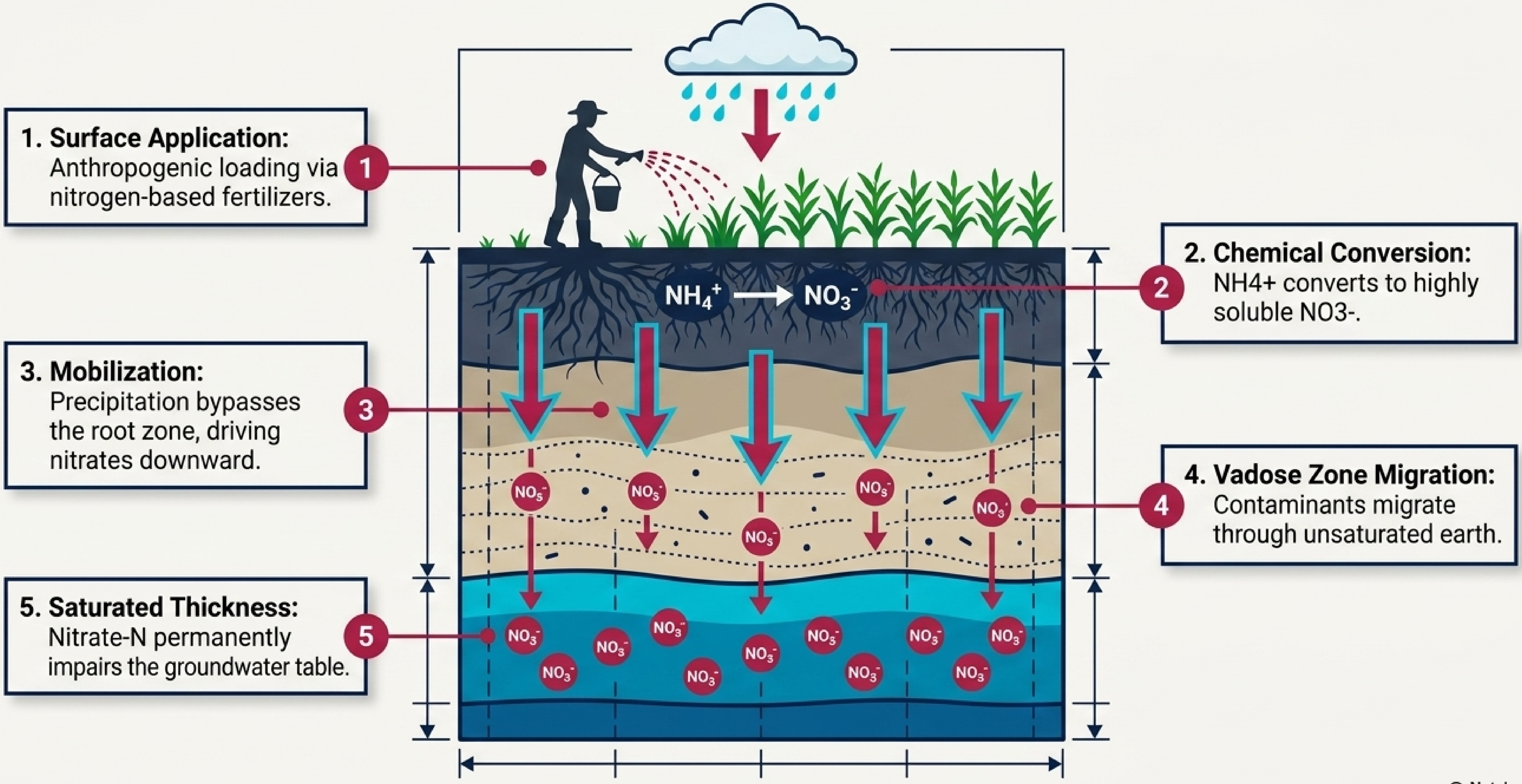
Millions rely on an unregulated and fragile subsurface resource





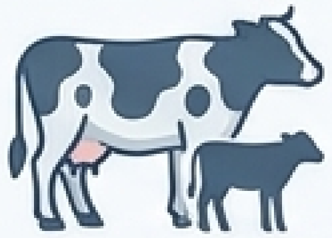
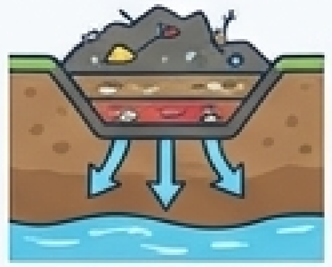
Insight: Because domestic wells are largely unregulated, the silent nature of nitrate contamination requires a transition from reactive testing to proactive identification.

The physical mechanics of agricultural leaching



NotebookLM

The Anatomy of Nitrate Pollution

	Source Type	Transport Mechanism	Impact Scale
	**Fertilizers**	Direct leaching of excess NO_3^-; conversion of NH_4^+ into mobile nitrate.	**Impact Scale:** Regional non-point source.
	**Manure**	Organic nitrogen mineralized to NO_3^-; non-point source leaching from livestock.	**Impact Scale:** Localized non-point source.
	**Industrial**	Point source discharge and chemical runoff leaching into the water table.	**Impact Scale:** Severe point source.
	**Landfill**	Pollution via leachate migration through compromised liners.	**Impact Scale:** Moderate point source.
	**Residential**	Leaking septic systems and turf-grass fertilizer leaching.	**Impact Scale:** Diffuse localized.

A hidden crisis crossing the legal limits of human consumption

10 mg/L NO3-N

The EPA maximum contaminant level (MCL) under the Safe Drinking Water Act.

Health Impacts

- Methemoglobinemia ('Blue Baby Syndrome')
- Birth defects & pregnancy complications
- Elevated cancer risks

82%

Percentage of domestic wells in the Bazile area estimated to be at risk of exceeding the 10 mg/L legal threshold.

A strategic transition from reactive testing to predictive strategy

Vulnerability Model

Multiple Logistic Regression (MLR)

- **Dataset:** Texas Gulf Coast
- **Output:** Calculates the Probability of Contamination
- **Benefit:** Eliminates cost-prohibitive physical testing

Economic Model

Life Cycle Costing (LCC)

- **Dataset:** Nebraska Case Study
- **Output:** Calculates Financial Reality of Remediation
- **Benefit:** Models impact over a 20-year horizon

Synthesis: Together, these models provide a high-precision framework to prioritize public health interventions and guide long-term socioeconomic policy.

The Geospatial Recipe: Modeling subsurface risk from surface data

GWDB Well Data

Texas Water Development Board dataset filtering high-risk observation points.

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NLCD Land Use

National Land Cover Database metrics capturing percent impervious surfaces.

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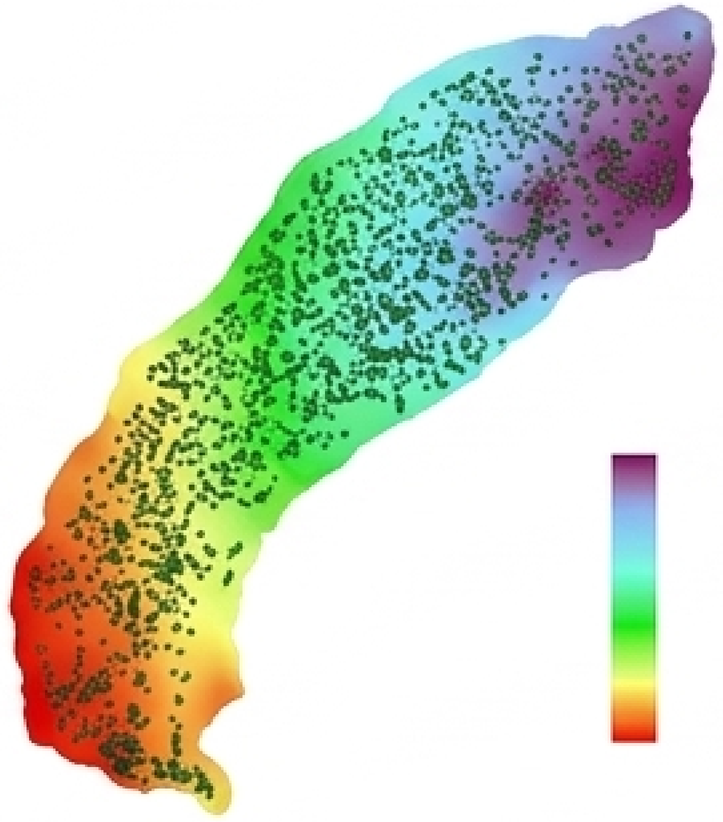
PRISM Precipitation



30-Year normal annual precipitation data.

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The MLR Vulnerability Model

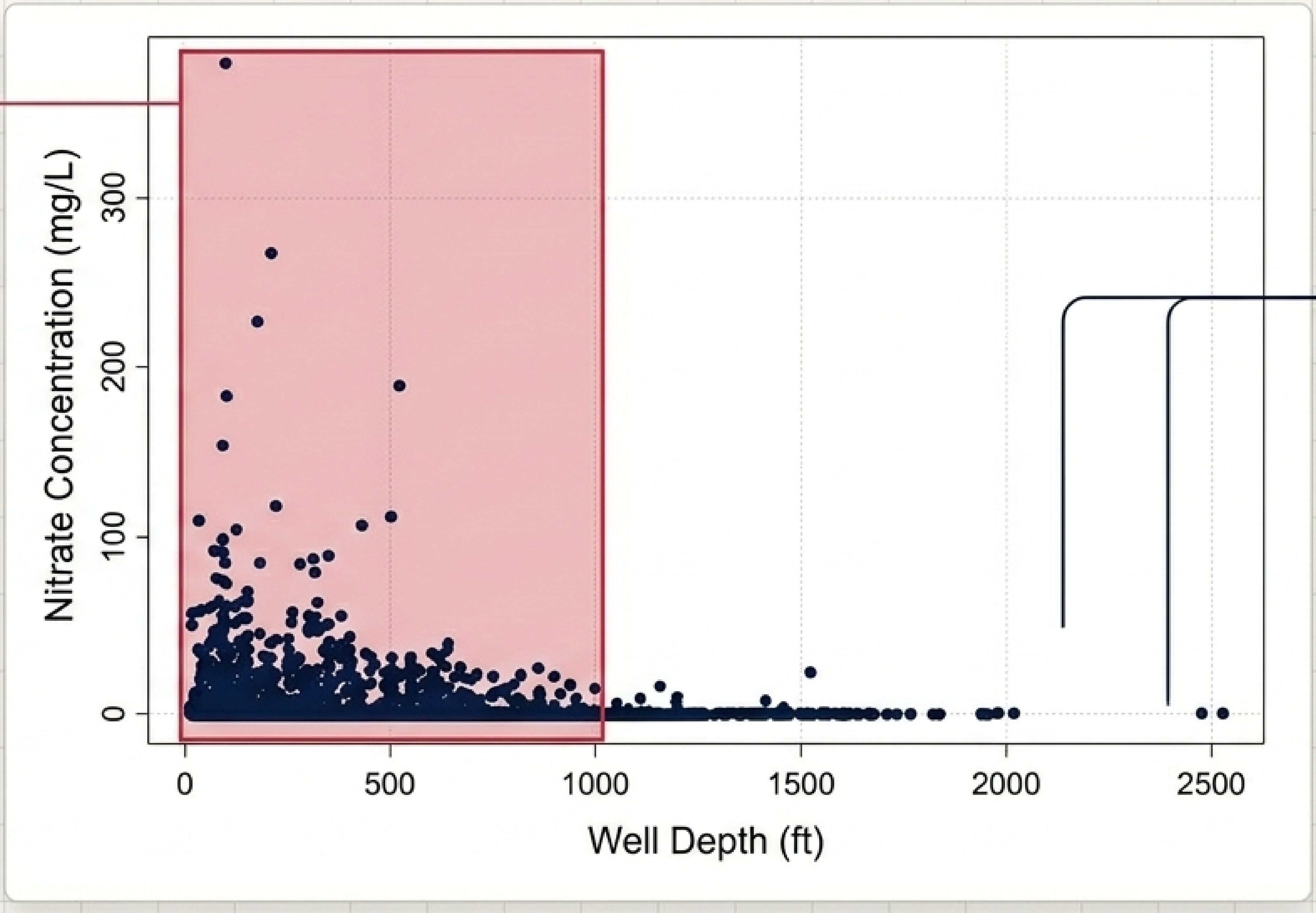
A synthesis of 12 distinct hydrogeological variables to map precise danger zones.

The 1,000-Foot Corridor: The primary pathway for anthropogenic loading

The Danger Zone

1,889 High-Risk Observations.

Wells shallower than 1,000 feet exhibit extreme vulnerability to surface-derived nitrates.



The Safe Zone

Deep wells remain protected by saturated thickness and geological buffering.

Strong diagnostic power eliminates the need for universal physical testing

Confidence Gauge

Confidence Gauge

Metric confirms high predictive precision.

What this metric means:

Statistical Validation

The Area Under the Receiver Operating Characteristic Curve (AUC) confirms high predictive precision.

Operational Impact

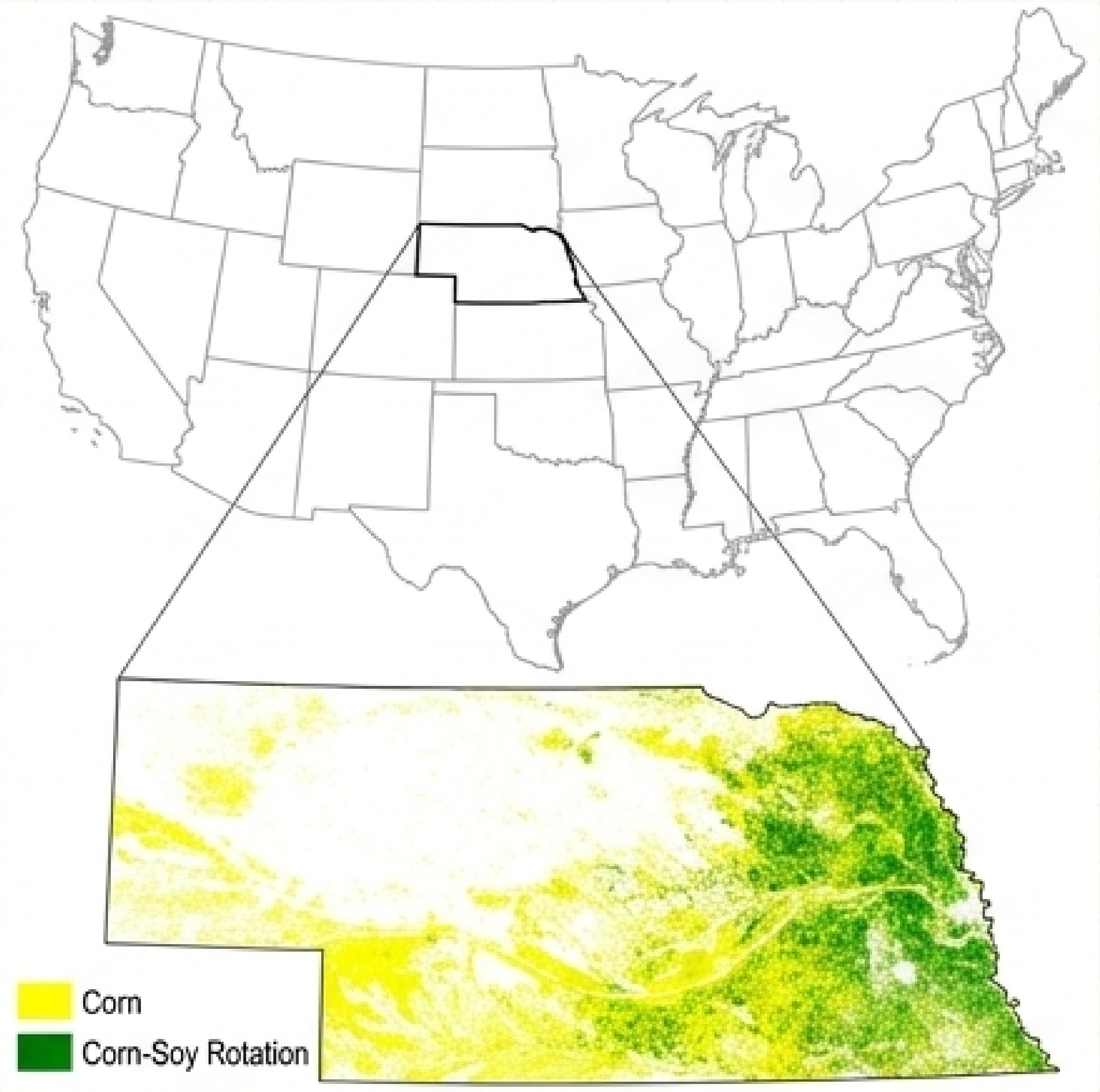
Accurately identifies vulnerable wellheads with an 75-25 train/test data split reliability.

Economic Benefit

Bypasses the exorbitant costs and logistical impossibilities of physically testing tens of thousands of rural domestic wells.

Quantifying the financial burden of remediation

The Nebraska Case Study



In the Tri-Basin area, 37% of wells are at risk of exceeding 7.5 mg/L, triggering immediate mitigation needs.

LCC Parameters

Capital Costs

Upfront investment for implementation and system installation.

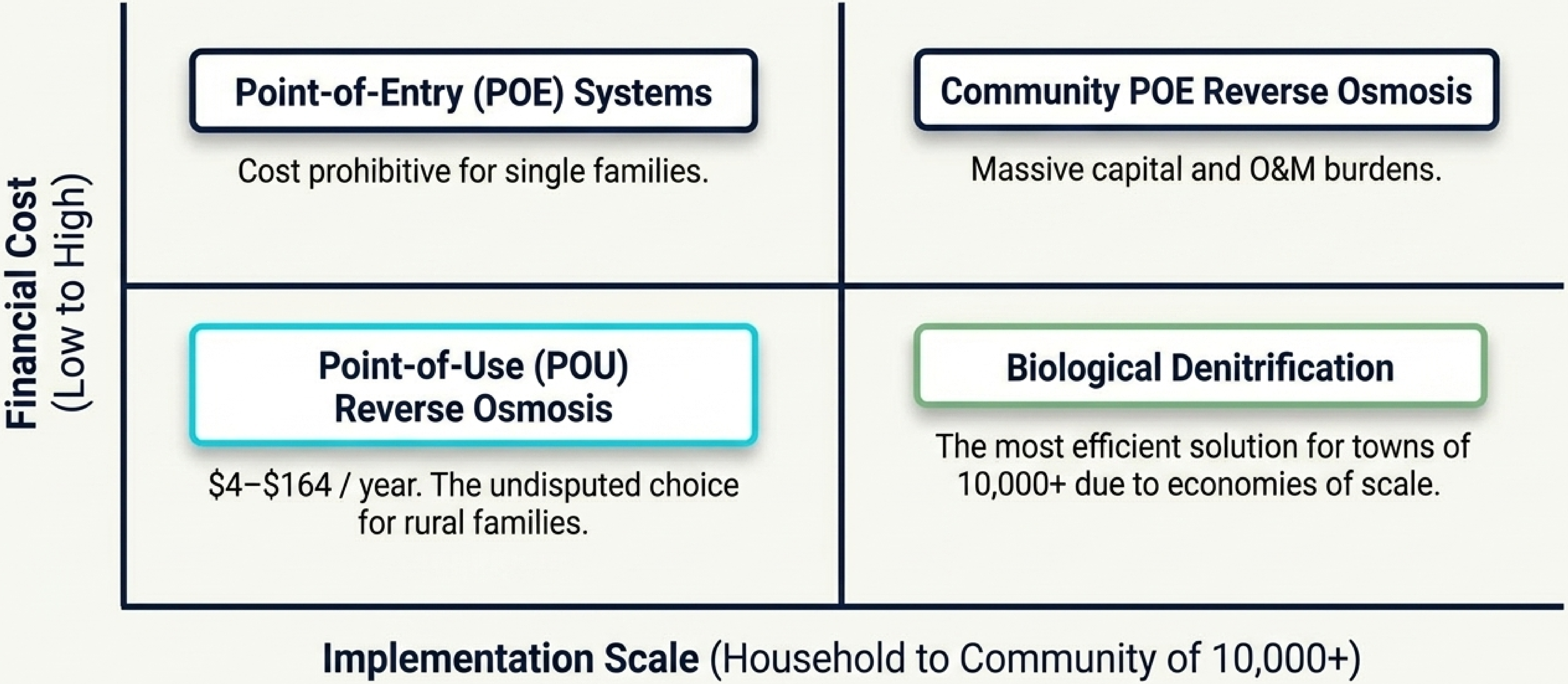
O&M Costs

Annual recurring costs for operation and maintenance.

Time Horizon

Analyzed over a 20-year lifecycle to capture the true socioeconomic burden.

Treatment viability is dictated entirely by economies of scale



The Remediation Cost Matrix

Technology	Capital Cost	O&M Cost	Implementation Speed	Ideal Scale
Point-of-Use (POU) Reverse Osmosis	Low	Moderate	Rapid	Individual households.
Ion Exchange & Distillation	Moderate	High Energy/O&M	Moderate	Secondary household alternative.
Point-of-Entry (POE) Reverse Osmosis	Extremely High	High	Slow	Rarely optimal.
Biological Denitrification	High	Low (At Scale)	Slow	Large community water systems (11 active plants in NE).

Breaking the cycle: A targeted socioeconomic policy recommendation

The high-cost trap of out-of-pocket mitigation.

Unregulated Contamination

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Physical Testing Delay

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Families buy expensive bottled water

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Health risks persist

Targeted assistance for low-income rural households.

MLR Predictive Identification

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State Funds Subsidize Capital Costs

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POU Reverse Osmosis Installed

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Safe drinking water secured for $4-$164/yr.

Integrated water management is not a luxury; it is a strategic necessity to protect rural populations from agricultural leaching.